

Poultry Disease Detection Using Machine Learning: A Review

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Abstract

Poultry farming plays an important role in ensuring food security and economic stability, especially in developing countries. However, the poultry industry continues to face important challenges due to the frequent outbreak of infectious diseases, resulting in significant economic losses and public health concerns. The recent progress in machine learning (ML) has shown great promise in the early detection of poultry diseases and increasing diagnosis, which enables timely intervention and control. This review paper presents a comprehensive observation of the existing ML approaches applied to the paper poultry disease detection, including supervised, unsupervised, and deep learning methods. It analyzes the functioning, dataset, and performance matrix employed in various studies, which highlights their strength, boundaries, and genuine world praise. Additionally, the paper identifies the current research interval and outlines the directions for the future to improve the systems of detection of the disease through the hybrid model, better dataset availability, and increased model accuracy. Review underlines ML's transformative ability to revolutionize poultry health management and support its adoption in accurate livestock farming.

Introduction

Poultry farming plays an important role in global food production, which supplies essential protein sources such as meat and eggs. In recent years, there has been a significant change in the region, which is developing in large -scale commercial enterprises from small backyard operations. This growth has been motivated by reproduction, nutrition, and improvement in agricultural management practices, which have greatly promoted production efficiency.

According to (Uzundumlu & Dilli, 2023), global chicken meat production was averaged of 45.93 million tons between 1961 and 2018. This production is expected to increase considerably, to reach about 139.19 million tons by 2025. According to (CAGR). By 2028, the market is estimated to increase by about \$ 494.55 billion, continuing at a CAGR of 6.4%. However, the intensity of poultry production has also increased the risk of disease outbreaks, which presents economic and public health challenges. Avian Influenza (AI), or bird flu, is a prominent example.

The Report of the Disease Control and Prevention Center (CDC, 2025) highlights the continuous spread of AI, which reported 67 human cases in the United States in 2024. Most individuals experienced mild symptoms, although one died in Louisiana in a serious case. Despite these incidents, the CDC says that the public health risk is low. Infectious Bermal Disease (IBD), also known as Gumboro disease, is another important disease. The disease is a highly infectious viral infection that mainly targets young chickens, weakens their immune systems, and makes them more susceptible to secondary infections. Due to its rapid transmission rate and high mortality, IBD in poultry can cause significant financial losses for farmers. These poultry underline the ongoing health hazards to monitor the effective disease and the immediate need for control strategies in farming. Early identification is necessary to apply timely intervention on time to reduce economic losses and prevent them from spreading further.

Artificial Intelligence (AI) and machine learning (ML) provide promising tools for disease detection and diagnosis in poultry. By analyzing data from sources such as temperature sensors, motion trackers, and vocalization monitors, these technologies can identify the signs of the disease before visual symptoms appear. This capacity allows for earlier treatment and control of the disease, which improves the results of recovery and reduces damage. Factors such as weather conditions and flock characteristics, such as ML techniques (Farahat et al., 2023), can also be included in the future model. With the help of advanced technology, it is possible to improve poultry health and productivity, reducing bird mortality. As

innovation continues, the strategies of disease prevention are expected to be more effective and accessible.

In summary, although poultry farming has made sufficient progress in improving productivity, management of the disease remains an important challenge. This review examines the role of machine learning in the detection of poultry disease by investigating current research, identifying ongoing issues, and discussing future opportunities. The study has mainly focused on the findings published in the last five years.

LITERATURE REVIEW

(Ehsan and Mohlipar, 2024) proposed a deeper learning structure of a real-time to analyze the behavior of broiler chickens in chicken houses free from the cage. Their system focuses on abnormal behavior, especially identifying inactivity and hobbling, indicating underlying health issues. The framework integrates three components: a deep learning model to detect chickens, a fast tracking module to monitor individuals checking in video frames, and an unusual behavior detection module. Experimental results demonstrated that the system could effectively assess herd behavior, allowing timely intervention and better health management in poultry farms.

A system developed by (Tran et al., 2024) is designed to identify flaws caused by transportation, processing or equipment failures. By taking advantage of the power of the transformer network, there is extreme visual features to increase the accuracy of detection of the CasclesFormer. Quantitative evaluation showed that the model outperform other state-of-the-art techniques in various tasks, including classification and division. Researchers highlighted the ability of the model to catch fine-and-fine characteristics, and they intend to release trained models and codebase to encourage cooperation.

In another work, by (Tong et al., 2024) developed a real-time health monitoring system using Edge Artificial Intelligence (Edge AI). The improved FCOS-light object detector employs knowledge distillation to maintain high accuracy by keeping the model compact. Using the mobile as a backbone introduces new advantages tasks to enhance both system classification and localization.

(V. et al., 2024) made a comprehensive review in poultry health and disease management, discovering the application of machine learning (ML) and deep learning (DL) techniques. The analysis covered studies published between 2010 and 2023, which focus on areas such as disease detection, behavior analysis, gender

identity, lame monitoring and weight prediction. The reviews examined various computational methods and challenges in the region, such as the need for a quality dataset, the complexity of convenience extraction, and limitations in the model lecturers. The authors also discussed the development of equipment and strategies used in poultry-centered AI applications.

(Cakic et al., 2023) examined the use of thermal imaging combined with artificial intelligence for the early detection of avian diseases. The study consisted of four groups of Ross 308 broilers, two of which were experimentally infected with Newcastle Disease (ND) or avian influenza (AI). Thermal images were captured at regular intervals and analyzed using MATLAB. The feature selection was done using a distance assessment method, and the support vector machines (SVMs) and Artificial Neural Network (ANN) were classified. The SVM-based approach obtained 97.2% and 100% accuracy in identifying AI and ND, respectively, within 24 hours of the infection, which exposed the ability of this method for rapid diagnosis.

In a study by (Saifullah et al., 2023), The Researchers applied the conventional neural network (CNN) models with transfer learning to detect fertility in chicken eggs without damaging the shells. They trained four models- VGG16, RESNET50, InceptionNET, and MobileNet on the dataset of 200 enhanced egg images. Of these, inceptionNET achieved the best performance with 98% accuracy, 100% sensitivity in detecting fertile eggs, and 96% specificity in identifying non-vigorous people. The study emphasized the utility of CNN-based transfer learning in improving hatchery practices and encouraged further research to refine these models.

(Machuve et al., 2022) compiled a large dataset of images labelled from both laboratory and agricultural settings. Several CNN architecture was tested, including VGG16, InceptionV3, MobileNetv2, and Xception. The fine-tuning model has improved significantly, with MobileNetv2 98.02% and Xception reaching 98.24%. The study concluded that mild models such as mobilonatav 2 are particularly suitable for deployment on-form due to their high accuracy and generalization capacity.

(Adebayo et al., 2023) developed an automatic system to detect digestive diseases in broiler chickens using deep CNN, Model-Faster R-CNN, and Yolo-V3- which were trained to classify unusual drops. Rapid R-CNN achieved better performance with a memory of 99.1% and an average accuracy of 93.3%, while Yolo-V3 showed a slightly lower results. This study demonstrated the viability of using image-based deep learning methods to detect rapid and accurate disease. At the end, finally,

(Campbell et al., 2019) created a machine vision-based monitoring system for broiler chickens by analyzing 2D asana shapes and mobility features. The system used an SVM classifier with a radial basis kernel function, with 97.5% accuracy. Despite the strong performance of the model, the authors recommended more verification in various chicken breeds and disease types to increase general suitability.

METHODOLOGY OF THE REVIEW

This review adopts a systematic approach to analyze recent progress in the application of machine learning (ML) to detect hen disease. This process followed several structured stages to ensure comprehensive coverage and relevance of materials reviewed. First, a diverse category of research articles was collected from major academic databases such as IEEE XPlore, Springer, Elsevier, and Google Scholar. The selection was limited to studies published within the last decade, to highlight the most current technological development, to pay special attention to the last five years. The Inclusion Criteria preferred articles that applied machine learning techniques, especially to poultry health monitoring and disease detection. After the article selection, the study was classified based on three main aspects: the type of data used (eg, pictures, audio, clinical records), planned machine learning models (e.g. CNNs, SVM, or ENSEMBLE methods), and evaluation metrics are used for performance model performance (eg, accuracy, accuracy, accuracy, accuracy, accuracy, recalish, recal To achieve insight into the technical method applicable in studies, a specific ML workflow was analyzed to detect poultry disease. These workflows commonly include the major stages seen:

- Dataset Development: Creation or collection of labelled datasets, often through image or sensor data acquisition from poultry farms or laboratory environments.
- Image Preprocessing: Standardization of input data through techniques such as noise reduction, normalization, cropping, and resizing to enhance model performance.
- Data Augmentation: Expansion of datasets using transformation techniques such as rotation, flipping, zooming, and scaling to improve model generalization and reduce overfitting.
- Feature Extraction and Selection: Extraction of relevant features from the data using manual or automated (e.g., deep learning) techniques, followed by selection of the most informative ones.

- **Image Classification:** Implementation of classification algorithms to categorize input data into various disease or health states, depending on the chosen model architecture.

This structured review methodology allowed for a detailed comparison of various approaches used in poultry disease detection and helped in identifying current trends, limitations, and research gaps in the field.

MACHINE LEARNING APPLICATIONS IN POULTRY DISEASE DETECTION

Various machine learning (ML) techniques have been applied to improve the accuracy and efficiency of disease detection in poultry farming. These applications are usually in three primary categories based on the type of data analyzed: visual data (picture), audio signal, and clinical or laboratory data.

The image-based detection conversion is one of the most widely used models to detect diseases in poultry through image analysis. They are particularly effective in identifying physical symptoms such as lesions, unnatural or unusual characteristics in chickens by learning spatial features from high-resolution images. These models have been used in the diagnosis of diseases such as cookidiosis, Newcastle disease, and avian influenza, analyzing fecal images, body conditions, and characteristics of carcasses.

The sound detection ML model has also been trained to analyze audio recordings for signs of respiratory crisis or behavioral abnormalities. These models can detect patterns in vocalization - such as cough, sneezing, or crisis call - which may indicate disease or discomfort. Although image-based systems are less common, sound-based monitoring is non-guspath and allows continuous observation without disturbing animals.

Clinical data analysis learning models such as Random Forest (RF), support vector machine (SVM), and logistic regression have been used to classify the disease conditions based on clinical symptoms and laboratory results. These models work by identifying correlations between known disease indicators (eg, body temperature, weight loss, blood markers) and confirming the diagnosis, which enables quick identification and future health evaluation.

In summary, in these applications, the ML model has shown high accuracy in identifying poultry diseases and often performs better by traditional clinical methods in terms of speed and stability. The option of ML technology depends largely on the type and quality of data available, as well as the unique identity of

goals. The image-based models remain the most effective due to the abundance of annotated visual datasets, while audio and clinical models provide supplementary equipment for extensive health monitoring in poultry farming.

COMPARATIVE ANALYSIS OF THE STUDIES

Comparative reviews of analyzed studies suggest that image-based machine learning models provide a higher level of accuracy and performance than other data types, such as audio or clinical data. This advantage is largely responsible for the abundance of annotated image datasets and the maturity of computer vision technologies in the domain of deep learning. In classification functions related to poultry disease detection, deep learning models, especially convolutional neural networks (CNN), support vector machines (SVMs), and random forests (RFS) consistently perform better than traditional machine learning methods. They excel in extracting hierarchical features from CNN images, allowing them to identify complex patterns and microscopic differences that may not be visible to the naked eye or are detected by simple algorithms. Transfer learning has also emerged as a powerful technique, especially in scenarios with limited training data. By taking advantage of the pre-trained model on a large-scale dataset, researchers are able to improve the model's performance and significantly reduce training time. Models such as VGG16, RESNET, and Mobilenet have shown strong results when they are well-tuned for poultry-specific tasks. However, it is important to note that the effectiveness of these models is highly dependent on the quality and variety of the dataset used. The factor can negatively affect the normalization capacity of the factor models, such as poor image resolutions, incompatible labelling, and limited class representation. In addition, while the image-based technology dominates the field, it comprises many data types, such as audio and environmental datasets, strengthening its strength, and can be able to detect more comprehensive diseases.

In conclusion, while deep learning models, especially trained on image data, have set new benchmarks in the detection of poultry disease, their success is closely associated with data quality, pre-processing methods, and the choice of model architecture. Future efforts should focus on the discovery of a multi-modal approach to standardize the dataset and pursue the area.

CHALLENGES AND LIMITATIONS IN ML-BASED POULTRY DISEASE DETECTION

Despite the remarkable progress in applying machine learning (ML) to detect poultry disease, there are many important challenges that limit its widely adoption, especially in real-world farm settings. Some of the limitations include:

Limited availability of high-quality datasets. This is one of the major obstacles is the lack of a large, well-annotated dataset. High-resolution images, labelled audio recording, and accurate clinical data collection can be resource-intensive and time-consuming. Additionally, the lack of standardized datasets in areas and poultry breeds restricts the generality of many models.

High computational demands of Deep learning models, especially high-dimensional data such as images or videos, often require adequate processing power and memory. These computational requirements make it difficult to deploy models with limited technical infrastructure or in low-resource areas where high-demonstration hardware may not be available.

Model interpretability. Many MLs and deep learning models serve as "black boxes", making it difficult for farmers, veterinarians, and other stakeholders to understand how decisions are made. This lack of transparency can obstruct confidence in AI-based recommendations and can make wrong predictions challenging or troubleshooting.

Moral and practical concerns, the integration of AI in animal health surveillance also raises moral concerns. To ensure that technology does not compromise animal welfare or interfere with the necessary human inspection. In addition, data privacy and ownership-especially when using farm-borne data-re-achieve unresolved issues in some areas.

Finally, while machine learning in poultry promises a lot in bringing revolution in health management, addressing these challenges is important for practical deployment. Creation of more inclusive datasets, optimizing models for edge computing, and an integrative AI framework are essential for reliability, purpose, and adoption.

RECOMMENDATION

As the area of detection of poultry disease is developing, many promising directions are emerging, which can further increase the effectiveness and access of machine learning (ML) applications in this domain.

1. Development of large and more diverse datasets. Future research should prioritize the creation of a broad dataset that includes various types of poultry breeds, environmental status and disease types. Such dataset models will improve generalization, reduce bias, and increase reliability in various agricultural settings and geographical areas.

2. Integration with IOT and Edge Computing Technologies. The Internet of Things (IOT) may be able to monitor in real-time and make decisions directly at the farm level by combining ML models with devices and edge computing platforms. This integration allows continuous health monitoring without relying on high-bandwidth networks or centralized computing resources, making the solutions more scalable and cost-effective.
3. Progress to explain AI (XAI). To increase confidence and purpose between end-users, there is a growing requirement for models that provide transparent and explanatory results. Explaining AI (XAI) techniques can help clarify how predictions are made, allowing farmers and veterinarians to better understand and function with the system recommendations.
4. Adaptation for small-scale and resource-limited farms. In poultry farming, several ML applications are designed for mass operations. Future research should detect mild, inexpensive, and user-friendly models in line with small farmers, especially in developing countries. These models should require a minimum infrastructure, providing accurate and timely insight.

In summary, the future of ML-based poultry disease detection is contained in creating inclusive, transparent and real-time systems that can work efficiently in diverse farming environments. Innovation in data collection, model design, and hardware integration will play an important role in shaping the next generation of intelligent poultry health monitoring solutions.

CONCLUSION

Machine learning has emerged as a transformative tool in the early detection and management of poultry diseases. By taking advantage of data from images, sounds, and clinical records, the ML models provide the ability to identify signs of disease more accurately and efficiently than many traditional methods. These technologies not only increase clinical accuracy but also intervene in time, which can significantly reduce mortality and improve overall herd productivity. Despite these advantages, many challenges such as limited data availability, computational demands, and model limitations should be addressed to enable the convenience adoption widely, especially in a low-resource farming environment. Additionally, moral ideas around automation in animal health surveillance call for careful implementation and regulation. Further integration of machine learning with real-time technologies like IOT, the development of more diverse datasets, and clear

progress in AI will be necessary to make these systems more effective, transparent, and accessible. With continuous innovation and targeted research, detection of ML-based poultry disease can revolutionize animal health management and contribute to the stability of the global poultry industry.

REFERENCE

- Adebayo, S., Aworinde, H. O., Akinwunmi, A. O., Alabi, O. M., Ayandiji, A., Sakpere, A. B., Adeyemo, A., Oyebamiji, A. K., Olaide, O., & Kizito, E. (2023). Enhancing poultry health management through machine learning-based analysis of vocalization signals dataset. *Data in Brief*, 50, 109528. <https://doi.org/10.1016/j.dib.2023.109528>
- Cakic, S., Popovic, T., Krco, S., Nedic, D., Babic, D., & Jovovic, I. (2023). Developing Edge AI Computer Vision for Smart Poultry Farms Using Deep Learning and HPC. *Sensors*, 23(6), 3002. <https://doi.org/10.3390/s23063002>
- Campbell, M., Miller, P., Díaz-Chito, K., Hong, X., McLaughlin, N., Parvinzamor, F., Martínez Del Rincón, J., & O'Connell, N. (2024). A computer vision approach to monitor activity in commercial broiler chickens using trajectory-based clustering analysis. *Computers and Electronics in Agriculture*, 217, 108591. <https://doi.org/10.1016/j.compag.2023.108591>
- Ehsan, T. Z., & Mohtavipour, S. M. (2024). *Broiler-Net: A Deep Convolutional Framework for Broiler Behavior Analysis in Poultry Houses* (arXiv:2401.12176). arXiv. <https://doi.org/10.48550/arXiv.2401.12176>
- Farahat, R. A., Khan, S. H., Rabaan, A. A., & Al-Tawfiq, J. A. (2023). The resurgence of Avian influenza and human infection: A brief outlook. *New Microbes and New Infections*, 53, 101122. <https://doi.org/10.1016/j.nmni.2023.101122>
- <https://www.cdc.gov/bird-flu/spotlights/h5n1-response-02262025.html>
- Machuve, D., Nwankwo, E., Mduma, N., & Mbelwa, J. (2022). Poultry diseases diagnostics models using deep learning. *Frontiers in Artificial Intelligence*, 5. <https://doi.org/10.3389/frai.2022.733345>
- Saifulah, S., Drezewski, R., Yudhana, A., Pranolo, A., Kaswijanti, W., Suryotomo, A. P., Putra, S. A., Khaliduzzaman, A., Prabuono, A. S., & Japkowicz, N. (2023). Nondestructive chicken egg fertility detection using CNN-transfer learning algorithms. *Jurnal Ilmiah Teknik Elektro Komputer Dan Informatika*, 9(3), 854–871. <https://doi.org/10.26555/jiteki.v9i3.26722>
- Tong, Q., Wang, J., Yang, W., Wu, S., Zhang, W., Sun, C., & Xu, K. (2024). Edge AI-Enabled Chicken Health Detection Based on Enhanced FCOS-Lite and Knowledge Distillation. *Computers and Electronics in Agriculture*, 226, 109432. <https://doi.org/10.1016/j.compag.2024.109432>
- Tran, M., Truong, S., Fernandes, A. F. A., Kidd, M. T., & Le, N. (2024). *CarcassFormer: An End-to-end Transformer-based Framework for Simultaneous Localization, Segmentation and Classification of Poultry Carcass Defect* (arXiv:2404.11429). arXiv. <https://doi.org/10.48550/arXiv.2404.11429>
- Uzundumlu, A. S., & Dilli, M. (2023). Estimating Chicken Meat Productions of Leader Countries for 2019-2025 Years. *Ciência Rural*, 53(2), e20210477. <https://doi.org/10.1590/0103-8478cr20210477>
- V., S., B. S., M., Laxmi, V., Kumar, A., & Shrivastava, S. (2024). Latest Trend and Challenges in Machine Learning- and Deep Learning-Based Computational Techniques in Poultry Health and Disease Management: A Review. *Journal of Computer Networks and Communications*, 2024(1), 8674250. <https://doi.org/10.1155/2024/8674250>